Beyond STRIDE: A Unified Multi-Attribute Framework for Trustworthy AI

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1 Introduction & Motivation

- **Global priority:** Ensuring trustworthiness as AI-based services operate with increasing autonomy
- **Gap in existing methods:** Traditional threat models (e.g., STRIDE) fail to handle AI-specific issues such as non-determinism, opacity, and data dependency

Research Questions(RQs):

- **RQ1.** Are traditional IT threat modeling methods sufficient for AI-based services?
- **RQ2.** What multi-attribute framework should be employed for trustworthy AI design?
- **RQ3.** What additional requirements are identified through the proposed framework?
- **RQ4.** Are the derived requirements aligned with regulatory obligations (e.g., EU AI Act) and consistently interpretable by external evaluators?

2 Insufficiency of Prior Threat and Risk Analysis Approaches for AI-Based Services RQ1

- **Up to 8 Attributes**: Leading frameworks (e.g., NIST AI RMF, EU AI Act) identify up to eight attributes that trustworthy AI systems should possess (Table 1).
- **Review of 40 Prior Studies:** Even the most comprehensive works cover at most 2/8 attributes, with the vast majority focusing solely on security via STRIDE or its variants.
- **Gap:** Even MAESTRO, a recent framework for agentic AI, remains security-focused and does not cover all eight attributes — motivating a unified multi-attribute framework

Table 1. Trustworthiness attributes in major frameworks and prior work coverage.

Framework	Sec	Saf	Rel	Tra	Acc	Fair	Hum	Priv
NIST AI RMF [1], Khlaaf (2023) [4]	✓	✓	✓	✓	✓	✓		✓
EU AI Act [2], ISO/IEC 22989 [3]	✓	✓	✓	✓	✓	✓	✓	✓
Japan AISI (2024) [5]	✓	✓		✓		✓	✓	✓
Max coverage of prior work	**2/8** (Mauri 2022 [6])							

Sec: Security; Saf: Safety; Rel: Reliability; Tra: Transparency; Acc: Accountability; Fair: Fairness; Hum: Human-centricity; Priv: Privacy.

3 Proposed Unified Multi-Attribute Framework RQ2

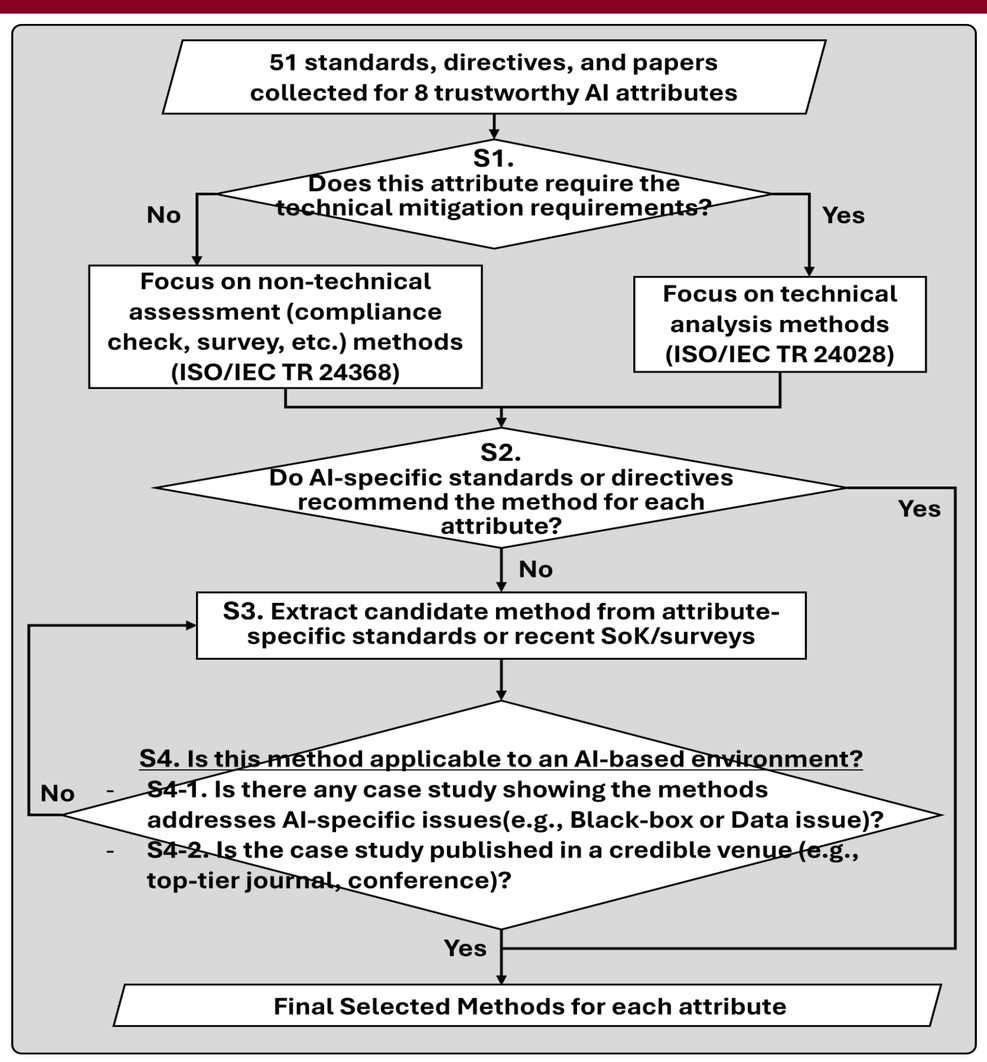

Figure 1. Four-stage filtering process (S1–S4).

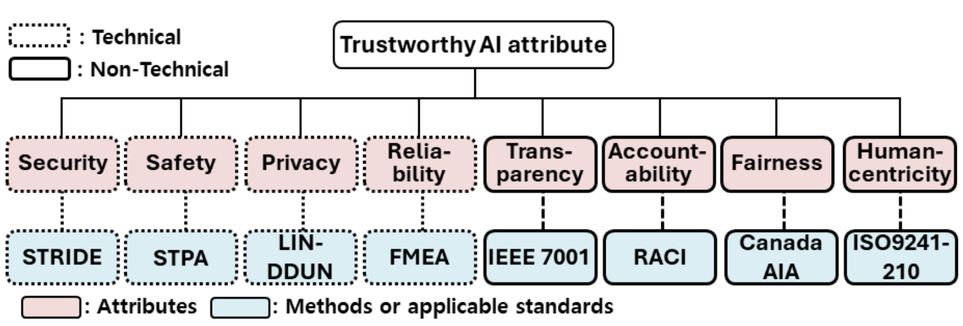

Figure 2. Eight analysis methods selected through Figure 1.

- Four-stage filtering process (S1–S4) reviewing 51 standards, directives, and papers to identify domain-specific analysis methods(Fig. 1)
- S1: Classify attributes → S2: Review AI standards/directives → S3: Review domain standards & SoK/surveys → S4: Check AI applicability and credibility → Result in Fig. 2
- Case Study: Deriving unified trustworthy AI requirements by applying Fig.3 to reference architecture of AI-based services → 103 unified requirements derived

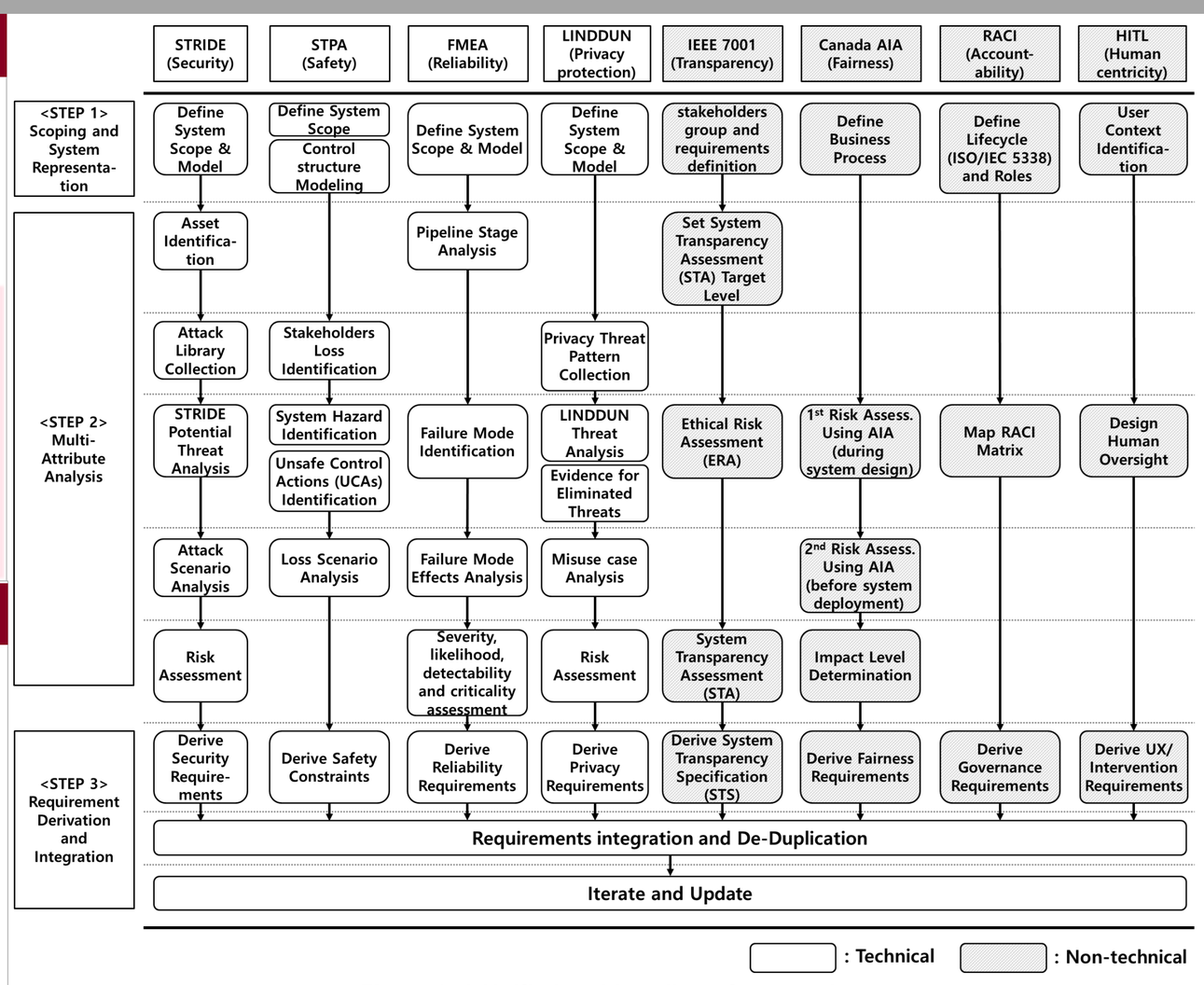

Figure 3. A Unified multi-attribute framework

4 What STRIDE Misses: Coverage Gap Analysis RQ3

- **STRIDE-only:** 27 reqs. — 24% EU AI Act coverage;
  ※ Zero coverage (0%) in Transparency (Art. 13) & Human Oversight (Art. 14).
- **Our Framework:** 103 requirements across all 8 attributes
  - e.g., STPA captured hazards from fully authorized AI actions — invisible in STRIDE

5 Regulatory Alignment and Real-World Validation RQ4

- **Regulatory Alignment:** 98% EU AI Act technical obligation coverage (vs. 24% STRIDE-only)
  * Detailed data is available on-line at https://github.com/HackProof/trustworthy_ai_by_design
- **Empirical Evaluation:** GPT-4o, Gemini 2.5 Flash, DeepSeek V3.1 — 92.2% inter-rater consistency
- **Real-World Validation:** 2nd / 47 teams at AI Red Teaming Challenge hosted by the MFDS (Ministry of Food and Drug Safety) — employing attack scenarios we identified

Table 2. EU AI Act coverage: STRIDE-only vs. multi-attribute.

EU AI Act Category	Units	STRIDE	Multi-Attr.
Art. 9 Risk management	18	4	18
Art. 10 Data governance	15	3	15
Art. 12 Logging & monitoring	8	4	8
Art. 13 Transparency	19	0	19
Art. 14 Human oversight	10	0	10
Art. 15 Technical robustness	20	12	19
Other obligations	10	1	9
Total	**100**	**24 (24%)**	**98 (98%)**

6 Conclusion

Security-focused threat modeling alone is insufficient for AI-based services. By integrating eight domain-specific methods, our framework demonstrates that trustworthy AI by design can be systematically achieved.

Acknowledgments

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